

## Annex 93A

(normative)

### Specification methods for electrical channels

#### 93A.1 Channel Operating Margin

The Channel Operating Margin (COM) is a figure of merit for a channel derived from a measurement of its scattering parameters. COM is related to the ratio of a calculated signal amplitude to a calculated noise amplitude as defined by Equation (93A–1).

$$COM = 20\log_{10}(A_s/A_{ni}) \quad (93A-1)$$

COM shall be calculated using the method described in this annex. The signal amplitude  $A_s$  is defined in 93A.1.6 and the noise amplitude  $A_{ni}$  is defined in 93A.1.7.

Figure 93A–1 illustrates the reference model that is the basis for the calculation for COM. The parameters used to calculate COM are listed in Table 93A–1. The values assigned to these parameters are defined by the Physical Layer specification that invokes the method (see Table 93A–2).

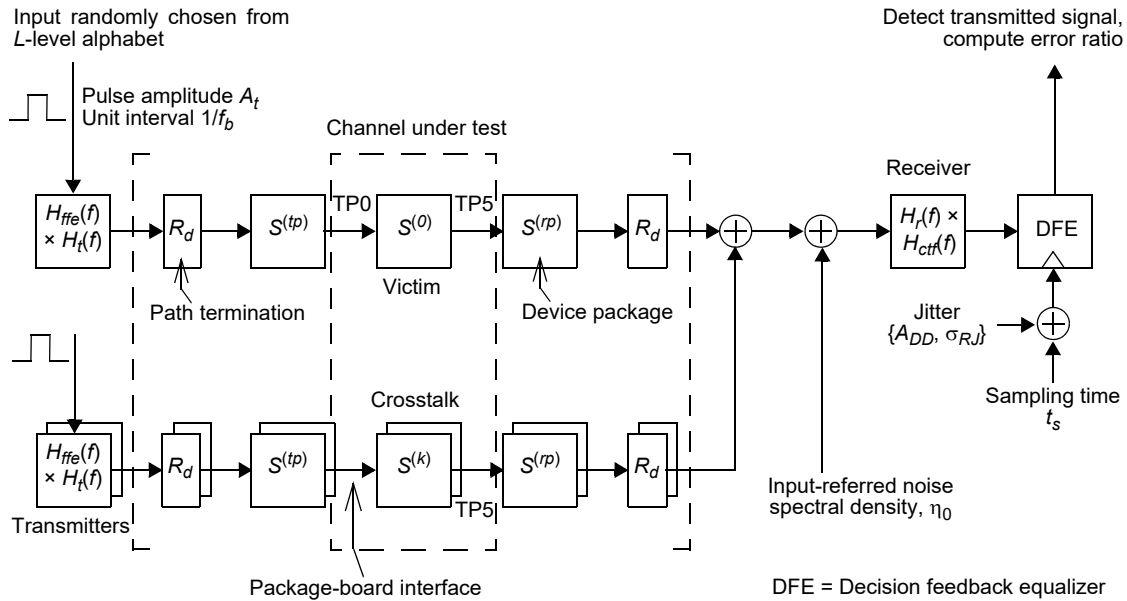


Figure 93A–1—COM reference model

**Table 93A-1—COM parameters**

| Parameter                                                                                                                                                                                                        | Reference | Symbol                           | Units                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------|----------------------------|
| Signaling rate                                                                                                                                                                                                   | 93A.1.1   | $f_b$                            | GBd                        |
| Maximum start frequency                                                                                                                                                                                          | 93A.1.1   | $f_{\min}$                       | GHz                        |
| Maximum frequency step                                                                                                                                                                                           | 93A.1.1   | $\Delta f$                       | GHz                        |
| Device package model<br>Single-ended device capacitance<br>Transmission line length<br>Single-ended package capacitance at package-to-board interface<br>Transmission line characteristic impedance <sup>a</sup> | 93A.1.2   | $C_d$<br>$z_p$<br>$C_p$<br>$Z_c$ | nF<br>mm<br>nF<br>$\Omega$ |
| Single-ended reference resistance                                                                                                                                                                                | 93A.1.2   | $R_0$                            | $\Omega$                   |
| Single-ended termination resistance                                                                                                                                                                              | 93A.1.3   | $R_d$                            | $\Omega$                   |
| Receiver 3 dB bandwidth                                                                                                                                                                                          | 93A.1.4.1 | $f_r$                            | GHz                        |
| Transmitter equalizer, minimum cursor coefficient                                                                                                                                                                | 93A.1.4.2 | $c(0)$                           | —                          |
| Transmitter equalizer, 2 <sup>nd</sup> pre-cursor coefficient <sup>b</sup><br>Minimum value<br>Maximum value<br>Step size                                                                                        | 93A.1.4.2 | $c(-2)$                          | —<br>—<br>—                |
| Transmitter equalizer, pre-cursor coefficient<br>Minimum value<br>Maximum value<br>Step size                                                                                                                     | 93A.1.4.2 | $c(-1)$                          | —<br>—<br>—                |
| Transmitter equalizer, post-cursor coefficient<br>Minimum value<br>Maximum value<br>Step size                                                                                                                    | 93A.1.4.2 | $c(1)$                           | —<br>—<br>—                |
| Continuous time filter, DC gain<br>Minimum value<br>Maximum value<br>Step size                                                                                                                                   | 93A.1.4.3 | $g_{DC}$                         | dB<br>dB<br>dB             |
| Continuous time filter, DC gain 2 <sup>c</sup><br>Minimum value<br>Maximum value<br>Step size                                                                                                                    | 93A.1.4.3 | $g_{DC2}$                        | dB<br>dB<br>dB             |
| Continuous time filter, zero frequency for $g_{DC} = 0$                                                                                                                                                          | 93A.1.4.3 | $f_z$                            | GHz                        |
| Continuous time filter, pole frequencies                                                                                                                                                                         | 93A.1.4.3 | $f_{p1}$<br>$f_{p2}$             | GHz                        |
| Continuous time filter, low frequency pole/zero <sup>d</sup>                                                                                                                                                     | 93A.1.4.3 | $f_{LF}$                         | GHz                        |
| Transmitter differential peak output voltage<br>Victim<br>Far-end aggressor<br>Near-end aggressor                                                                                                                | 93A.1.5   | $A_v$<br>$A_{fe}$<br>$A_{ne}$    | V<br>V<br>V                |
| Number of signal levels                                                                                                                                                                                          | 93A.1.6   | $L$                              | —                          |
| Level separation mismatch ratio                                                                                                                                                                                  | 93A.1.6   | $R_{LM}$                         | —                          |
| Transmitter signal-to-noise ratio                                                                                                                                                                                | 93A.1.6   | $SNR_{TX}$                       | dB                         |

**Table 93A-1—COM parameters (*continued*)**

| Parameter                                  | Reference | Symbol        | Units               |
|--------------------------------------------|-----------|---------------|---------------------|
| Number of samples per unit interval        | 93A.1.6   | $M$           | —                   |
| Decision feedback equalizer (DFE) length   | 93A.1.6   | $N_b$         | UI                  |
| Normalized DFE coefficient magnitude limit | 93A.1.6   | $b_{\max}(n)$ | —                   |
| Random jitter, RMS                         | 93A.1.6   | $\sigma_{RJ}$ | UI                  |
| Dual-Dirac jitter, peak                    | 93A.1.6   | $A_{DD}$      | UI                  |
| One-sided noise spectral density           | 93A.1.6   | $\eta_0$      | V <sup>2</sup> /GHz |
| Target detector error ratio                | 93A.1.7   | $DER_0$       | —                   |

<sup>a</sup>Some clauses that invoke this method do not provide a value for  $Z_c$ . See 93A.1.2.

<sup>b</sup>Some clauses that invoke this method do not provide a value for  $c(-2)$ . See 93A.1.6.

<sup>c</sup>Some clauses that invoke this method do not provide a value for  $g_{DC2}$ . See 93A.1.4.3.

<sup>d</sup>Some clauses that invoke this method do not provide a value for  $f_{LF}$ . See 93A.1.4.3.

**Table 93A-2—Physical Layer specifications that employ COM**

| Physical Layer             | Parameter values |
|----------------------------|------------------|
| 25GBASE-CR (Clause 110)    | Table 110-11     |
| 25GBASE-CR-S (Clause 110)  | Table 110-11     |
| 25GBASE-KR (Clause 111)    | Table 111-8      |
| 25GBASE-KR-S (Clause 111)  | Table 111-8      |
| 25GAUI C2C (Annex 109A)    | Table 83D-6      |
| 50GBASE-CR (Clause 136)    | Table 136-18     |
| 50GBASE-KR (Clause 137)    | Table 137-6      |
| LAUI-2 C2C (Annex 135B)    | Table 83D-6      |
| 50GAUI-2 C2C (Annex 135D)  | Table 83D-6      |
| 50GAUI-1 C2C (Annex 135F)  | Table 120D-8     |
| 100GBASE-CR4 (Clause 92)   | Table 93-8       |
| 100GBASE-CR2 (Clause 136)  | Table 136-18     |
| 100GBASE-KR4 (Clause 93)   | Table 93-8       |
| 100GBASE-KP4 (Clause 94)   | Table 94-17      |
| 100GBASE-KR2 (Clause 137)  | Table 137-6      |
| CAUI-4 C2C (Annex 83D)     | Table 83D-6      |
| 100GAUI-4 C2C (Annex 135D) | Table 83D-6      |
| 100GAUI-2 C2C (Annex 135F) | Table 120D-8     |
| 200GBASE-CR4 (Clause 136)  | Table 136-18     |
| 200GBASE-KR4 (Clause 137)  | Table 137-6      |

**Table 93A–2—Physical Layer specifications that employ COM**

| Physical Layer          | Parameter values         |
|-------------------------|--------------------------|
| 200GAUI-8 (Annex 120B)  | Table 83D–6 <sup>a</sup> |
| 200GAUI-4 (Annex 120D)  | Table 120D–8             |
| 400GAUI-16 (Annex 120B) | Table 83D–6 <sup>a</sup> |
| 400GAUI-8 (Annex 120D)  | Table 120D–8             |

<sup>a</sup>With the exceptions given in 120B.4.

### 93A.1.1 Measurement of the channel

The channel consists of a victim signal path plus some number of far-end and near-end crosstalk paths. The total number of paths for a given channel is denoted as  $K$  and, by convention, the path index  $k=0$  corresponds to the victim path. The number of crosstalk paths is a function of the structure of the system. All significant contributors to the channel crosstalk should be included in the calculation of COM.

Each signal path is represented by a set of frequency-dependent scattering parameters. For the purpose of the calculation of COM, references to scattering parameters correspond to the differential-mode scattering parameters. The scattering parameters measured at frequency  $f$  are presented as the  $2 \times 2$  matrix  $S(f)$  as defined by Equation (93A–2).

$$S(f) = \begin{bmatrix} s_{11}(f) & s_{12}(f) \\ s_{21}(f) & s_{22}(f) \end{bmatrix} \quad (93A-2)$$

The relationship between  $S(f)$  and other commonly cited characteristics is as follows. The insertion loss is the magnitude in dB of either  $1/s_{12}(f)$  or  $1/s_{21}(f)$ . The input and output return loss are the magnitude in dB of  $1/s_{11}(f)$  and  $1/s_{22}(f)$ , respectively.

The scattering parameters for the victim signal path are measured from TP0 to TP5. The scattering parameters for each crosstalk path are measured from the package-to-board interface of the aggressor transmitter to TP5. The frequency-dependent scattering matrix for signal path  $k$  is denoted as  $S^{(k)}(f)$ . The reference impedance for scattering parameter measurements is  $100 \Omega$ .

It is recommended that the scattering parameters be measured with uniform frequency step no larger than  $\Delta f$  from a start frequency no larger than  $f_{\min}$  to a stop frequency of at least the signaling rate  $f_b$ .

### 93A.1.2 Transmitter and receiver device package models

Each signal path in the channel is augmented to reflect the likely influence of transmitter and receiver device packages. The device package models are two-port networks defined by their scattering parameters. The scattering parameters are calculated using the method defined in 93A.1.2.1 through 93A.1.2.4.

Each signal path in the channel is represented by the scattering matrix  $S^{(k)}$ . The augmented signal path is denoted as  $S_p^{(k)}$  and is defined by Equation (93A–3).

$$S_p^{(k)} = \text{cascade}(\text{cascade}(S^{(tp)}, S^{(k)}), S^{(rp)}) \quad (93A-3)$$

The function `cascade()` is defined in 93A.1.2.1.  $S^{(tp)}$  and  $S^{(rp)}$  are defined in 93A.1.2.4. If  $k$  corresponds to a near-end crosstalk path,  $S^{(tp)}$  is calculated with the smallest value of  $z_p$  specified by the clause that invokes this method.

### 93A.1.2.1 Cascade connection of two-port networks

The connection of a pair of two-port networks  $x$  and  $y$  such that port 2 of network  $x$  is connected to port 1 of network  $y$  may be represented by an equivalent two-port network  $z$ . Port 1 of network  $z$  corresponds to port 1 of network  $x$  and port 2 network  $z$  corresponds to port 2 of network  $y$ . The scattering parameters of network  $z$  are given in terms of the scattering parameters of networks  $x$  and  $y$  by Equation (93A-4) through Equation (93A-7).

$$s_{11}^{(z)} = s_{11}^{(x)} + \frac{s_{21}^{(x)} s_{11}^{(y)} s_{12}^{(x)}}{1 - s_{22}^{(x)} s_{11}^{(y)}} \quad (93A-4)$$

$$s_{12}^{(z)} = \frac{s_{12}^{(x)} s_{12}^{(y)}}{1 - s_{22}^{(x)} s_{11}^{(y)}} \quad (93A-5)$$

$$s_{21}^{(z)} = \frac{s_{21}^{(x)} s_{21}^{(y)}}{1 - s_{22}^{(x)} s_{11}^{(y)}} \quad (93A-6)$$

$$s_{22}^{(z)} = s_{22}^{(y)} + \frac{s_{12}^{(y)} s_{22}^{(x)} s_{21}^{(y)}}{1 - s_{22}^{(x)} s_{11}^{(y)}} \quad (93A-7)$$

For the purpose of this annex, this set of operations is referred to using the shorthand notation  $S^{(z)} = \text{cascade}(S^{(x)}, S^{(y)})$ .

### 93A.1.2.2 Two-port network for a shunt capacitance

The scattering parameters for a shunt capacitance with value  $C$  are defined by Equation (93A-8) where  $j = \sqrt{-1}$  and  $\omega = 2\pi f$ .

$$S(C) = \frac{1}{2 + j\omega CR_0} \begin{bmatrix} -j\omega CR_0 & 2 \\ 2 & -j\omega CR_0 \end{bmatrix} \quad (93A-8)$$

The scattering parameters for the device capacitance  $C_d$  are denoted as  $S^{(d)} = S(C_d)$  and the scattering parameters for the board capacitance  $C_p$  are denoted as  $S^{(p)} = S(C_p)$ .

### 93A.1.2.3 Two-port network for the package transmission line

The scattering parameters for the package transmission line model are a function of the complex propagation coefficient defined by Equation (93A-9), Equation (93A-10), and Equation (93A-11) and the reflection coefficient defined by Equation (93A-12). The values of the parameters that appear in these equations are defined in Table 93A-3. Where a value for  $Z_c$  is not provided by the clause that invokes this method, it takes the value 78.2  $\Omega$ . The units of  $f$  are GHz.

$$\gamma(f) = \begin{cases} \gamma_0 & f = 0 \\ \gamma_0 + \gamma_1 \sqrt{f} + \gamma_2(f)f & f > 0 \end{cases} \quad (93A-9)$$

$$\gamma_1 = a_1(1+j) \quad (93A-10)$$

$$\gamma_2(f) = a_2(1 - j(2/\pi)\log_e(f/1 \text{ GHz})) + j2\pi\tau \quad (93A-11)$$

$$\rho = \frac{Z_c - 2R_0}{Z_c + 2R_0} \quad (93A-12)$$

**Table 93A-3—Transmission line model parameters and values**

| Parameter  | Value                  | Units                 |
|------------|------------------------|-----------------------|
| $\gamma_0$ | 0                      | 1/mm                  |
| $a_1$      | $1.734 \times 10^{-3}$ | ns <sup>1/2</sup> /mm |
| $a_2$      | $1.455 \times 10^{-4}$ | ns/mm                 |
| $\tau$     | $6.141 \times 10^{-3}$ | ns/mm                 |

The scattering parameters for a package transmission line of length  $z_p$  are defined by Equation (93A-13) and Equation (93A-14). The units of  $z_p$  are mm.

$$s_{11}^{(l)}(f) = s_{22}^{(l)}(f) = \frac{\rho(1 - \exp(-\gamma(f)2z_p))}{1 - \rho^2 \exp(-\gamma(f)2z_p)} \quad (93A-13)$$

$$s_{21}^{(l)}(f) = s_{12}^{(l)}(f) = \frac{(1 - \rho^2)\exp(-\gamma(f)z_p)}{1 - \rho^2 \exp(-\gamma(f)2z_p)} \quad (93A-14)$$

The transmission line scattering parameter matrix is then denoted as  $S^{(l)}$ .

#### 93A.1.2.4 Assembly of transmitter and receiver device package models

The scattering parameters for the transmitter device package model  $S^{(tp)}$  are the result of the cascade connection of the device capacitance, package transmission line, and board capacitance as defined by Equation (93A-15).

$$S^{(tp)} = \text{cascade}(\text{cascade}(S^{(d)}, S^{(l)}), S^{(p)}) \quad (93A-15)$$

Similarly, the scattering parameters for the receiver device package model  $S^{(rp)}$  are the result of the cascade connection of the board capacitance, package transmission line, and device capacitance as defined by Equation (93A-16).

$$S^{(rp)} = \text{cascade}(\text{cascade}(S^{(p)}, S^{(l)}), S^{(d)}) \quad (93A-16)$$

### 93A.1.3 Path terminations

The input to each signal path is terminated by an impedance defined by the reflection coefficient  $\Gamma_1$ . The output of each signal path is terminated by an impedance defined by the reflection coefficient  $\Gamma_2$ .

The reflection coefficients  $\Gamma_1$  and  $\Gamma_2$  are defined by Equation (93A-17).

$$\Gamma_1 = \Gamma_2 = \frac{R_d - R_0}{R_d + R_0} \quad (93A-17)$$

The voltage transfer function of the terminated signal path is defined by Equation (93A-18) where  $\Delta S(f) = s_{11}(f)s_{22}(f) - s_{12}(f)s_{21}(f)$ .

$$H_{21}(f) = \frac{s_{21}(f)(1 - \Gamma_1)(1 + \Gamma_2)}{1 - s_{11}(f)\Gamma_1 - s_{22}(f)\Gamma_2 + \Gamma_1\Gamma_2\Delta S(f)} \quad (93A-18)$$

The voltage transfer function for the signal path represented by  $S_p^{(k)}(f)$  is denoted  $H_{21}^{(k)}(f)$ .

### 93A.1.4 Filters

The voltage transfer function for each signal path  $H_{21}^{(k)}(f)$  (see 93A.1.3) is multiplied by a set of filter transfer functions to yield  $H^{(k)}(f)$  as shown in Equation (93A-19).

$$H^{(k)}(f) = H_{ffe}(f)H_t(f)H_{21}^{(k)}(f)H_r(f)H_{ctf}(f) \quad (93A-19)$$

The receiver noise filter  $H_r(f)$  is defined in 93A.1.4.1, the transmitter transition time filter  $H_t(f)$  is defined by Equation (93A-46) where  $T_r$  is 0 unless defined otherwise for the Physical Layer specification that invokes this method, the transmitter equalizer  $H_{ffe}(f)$  is defined in 93A.1.4.2, and the receiver equalizer  $H_{ctf}(f)$  is defined in 93A.1.4.3.

The filtered voltage transfer function  $H^{(k)}(f)$  is used to compute the pulse response (see 93A.1.5).

#### 93A.1.4.1 Receiver noise filter

$H_r(f)$  is a noise filter defined by Equation (93A-20).

$$H_r(f) = \frac{1}{1 - 3.414214(f/f_r)^2 + (f/f_r)^4 + j2.613126(f/f_r - (f/f_r)^3)} \quad (93A-20)$$

#### 93A.1.4.2 Transmitter equalizer

$H_{ffe}(f)$  is defined by Equation (93A-21) and is intended to represent the transmitter equalizer. If  $k$  corresponds to a near-end crosstalk path, then  $c(-2)$ ,  $c(-1)$ , and  $c(1)$  are all zero regardless of the values used for the other paths. The value of the “cursor” coefficient  $c(0)$  is set to  $1 - |c(-2)| - |c(-1)| - |c(1)|$  for any value of  $c(-2)$ ,  $c(-1)$ , and  $c(1)$ . If the value of  $c(0)$  is less than the specified minimum value, the corresponding combination of  $c(-2)$ ,  $c(-1)$ , and  $c(1)$  is considered invalid and is not used to calculate COM.

$$H_{ffe}(f) = \sum_{i=-2}^1 c(i) \exp\left(-j2\pi(i+2)\frac{f}{f_b}\right) \quad (93A-21)$$

### 93A.1.4.3 Receiver equalizer

$H_{ctf}(f)$  is defined by Equation (93A–22). Where values for  $g_{DC2}$  and  $f_{LF}$  are not provided by the clause that invokes this method,  $g_{DC2}$  takes the value 0 and  $f_{LF}$  takes the value 1.

$$H_{ctf}(f) = \frac{\left(10^{\frac{g_{DC}}{20}} + j\frac{f}{f_z}\right)\left(10^{\frac{g_{DC2}}{20}} + j\frac{f}{f_{LF}}\right)}{\left(1 + j\frac{f}{f_{p1}}\right)\left(1 + j\frac{f}{f_{p2}}\right)\left(1 + j\frac{f}{f_{LF}}\right)} \quad (93A-22)$$

### 93A.1.5 Pulse response

The pulse response of a signal path is defined to be the output of the path following the application of a rectangular pulse one unit interval in duration at its input. First define the function  $X(f)$  per Equation (93A–23) where  $\text{sinc}(x) = \sin(\pi x)/(\pi x)$  and  $T_b = 1/f_b$  is the unit interval.

$$X(f) = A_t T_b \text{sinc}(f T_b) \quad (93A-23)$$

$X(f)$  is a function of  $A_t$ , which in turn is based on the path index  $k$ . If  $k=0$ , i.e., the victim path, then  $A_t = A_v$ . If  $k$  corresponds to a far-end crosstalk path, then  $A_t = A_{fe}$ . If  $k$  corresponds to a near-end crosstalk path, then  $A_t = A_{ne}$ .

The pulse response  $h^{(k)}(t)$  is derived from the voltage transfer function  $H^{(k)}(f)$  (see 93A.1.4) using Equation (93A–24).

$$h^{(k)}(t) = \int_{-\infty}^{\infty} X(f) H^{(k)}(f) \exp(j2\pi ft) df \quad (93A-24)$$

NOTE 1—COM is expected to be computed from measurements at discrete frequencies that cover a limited span (see 93A.1.1). The inverse Fourier transform depicted in Equation (93A–24) is likely to be implemented as a discrete Fourier transform and the filtered voltage transfer function may need to be extrapolated (both to DC and to one half of the sampling frequency) for this computation. The extrapolation method and sampling frequency should be chosen carefully to limit the error in the COM computation.

NOTE 2—The time span of the pulse response in unit intervals,  $N$ , is limited in practice by frequency step  $\Delta f (N = f_b/\Delta f)$  but in general should be set to include all significant components of the pulse response.

### 93A.1.6 Determination of variable equalizer parameters

COM is a function of the variables  $c(-2)$ ,  $c(-1)$ ,  $c(1)$ ,  $g_{DC}$ , and  $g_{DC2}$ . The following procedure is used to determine the values of these variables that are used to calculate COM.

- Compute the pulse response  $h^{(k)}(t)$  of each signal path  $k$  for a given  $c(-2)$ ,  $c(-1)$ ,  $c(1)$ ,  $g_{DC}$ , and  $g_{DC2}$  using the procedure defined in 93A.1.5.
- Define  $t_s$  to be the time that satisfies Equation (93A–25). If there are multiple values of  $t_s$  that satisfy the equation, then the first value prior to the peak of  $h^{(0)}(t)$  is selected. The coefficients of the decision feedback equalizer  $b(n)$  are computed as shown in Equation (93A–26). If  $N_b$  is 0, then the  $b(n)$  is considered to be zero for all  $n$ .
- Define  $A_s$  to be  $R_{LM} h^{(0)}(t_s)/(L-1)$ .
- Compute  $\sigma_{TX}^2$  per Equation (93A–30) and Equation (93A–29). This represents the noise output from the transmitter.

- e) Compute  $h_{ISI}(n)$  per Equation (93A–27). This represents the residual intersymbol interference (ISI) after decision feedback equalization. The corresponding ISI amplitude variance  $\sigma_{ISI}^2$  is computed per Equation (93A–31) and Equation (93A–29).
- f) Compute the slope of the pulse response of the victim path  $h_J(n)$  as shown in Equation (93A–28). The variance of the amplitude error due to timing jitter  $\sigma_J^2$  is computed per Equation (93A–32) and Equation (93A–29).
- g) The variance of the amplitude for path  $k$  is given by Equation (93A–33) where the phase index  $m$  can assume any integer value from 0 to  $M-1$ . Denote the value of  $m$  that maximizes the variance for path  $k$  as  $i$ . The variance of the amplitude for the combination of all crosstalk paths  $\sigma_{XT}^2$  is then computed using Equation (93A–34), which is the sum of the maximum variances for the individual paths  $k=1$  to  $K-1$ .
- h) Compute the variance of the noise at the output of the receive equalizer  $\sigma_N^2$  based on the one-sided spectral density  $\eta_0$  referred to the receiver noise filter input per Equation (93A–35).
- i) Compute the figure of merit (FOM) per Equation (93A–36).

$$h^{(0)}(t_s - T_b) = h^{(0)}(t_s + T_b) - h^{(0)}(t_s)b(1) \quad (93A-25)$$

$$b(n) = \left\{ \begin{array}{ll} -b_{\max}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) < -b_{\max}(n) \\ b_{\max}(n) & h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) > b_{\max}(n) \\ h^{(0)}(t_s + nT_b)/h^{(0)}(t_s) & \text{otherwise} \end{array} \right\} \quad (93A-26)$$

$$h_{ISI}(n) = \left\{ \begin{array}{ll} 0 & n = 0 \\ h^{(0)}(t_s + nT_b) - h^{(0)}(t_s)b(n) & 1 \leq n \leq N_b \\ h^{(0)}(t_s + nT_b) & \text{otherwise} \end{array} \right\} \quad (93A-27)$$

$$h_J(n) = \frac{h^{(0)}(t_s + (n + 1/M)T_b) - h^{(0)}(t_s + (n - 1/M)T_b)}{2/M} \quad (93A-28)$$

$$\sigma_X^2 = \frac{L^2 - 1}{3(L - 1)^2} \quad (93A-29)$$

$$\sigma_{TX}^2 = [h^{(0)}(t_s)]^2 10^{-SNR_{TX}/10} \quad (93A-30)$$

$$\sigma_{ISI}^2 = \sigma_X^2 \sum_n h_{ISI}^2(n) \quad (93A-31)$$

$$\sigma_J^2 = (A_{DD}^2 + \sigma_{RJ}^2) \sigma_X^2 \sum_n h_J^2(n) \quad (93A-32)$$

$$[\sigma_m^{(k)}]^2 = \sigma_X^2 \sum_n [h^{(k)}((m/M + n)T_b)]^2 \quad (93A-33)$$

$$\sigma_{XT}^2 = \sum_{k=1}^{K-1} [\sigma_i^{(k)}]^2 \quad (93A-34)$$

$$\sigma_N^2 = \eta_0 \int_0^\infty |H_r(f)H_{ctf}(f)|^2 df \quad (93A-35)$$

$$FOM = 10 \log_{10} \left( \frac{A_s^2}{\sigma_{TX}^2 + \sigma_{ISI}^2 + \sigma_J^2 + \sigma_{XT}^2 + \sigma_N^2} \right) \quad (93A-36)$$

The FOM is calculated for each permitted combination of  $c(-2)$ ,  $c(-1)$ ,  $c(1)$ ,  $g_{DC}$ , and  $g_{DC2}$  values per Table 93A-1, where any parameters not provided by the clause that invokes this method are set to 0. The combination of values that maximizes the FOM, including the corresponding value of  $t_s$ , is used for the calculation of the interference and noise amplitude in 93A.1.7 and the calculation of COM in 93A.1.

### 93A.1.7 Interference and noise amplitude

Given the values of  $c(-1)$ ,  $c(1)$ ,  $g_{DC}$ ,  $g_{DC2}$ , and  $t_s$  derived in 93A.1.6, compute the combined interference and noise distribution  $p(y)$  per 93A.1.7.3. The corresponding cumulative distribution function is  $P(y)$  as defined by Equation (93A-37).

$$P(y) = \int_{-\infty}^y p(y) dy \quad (93A-37)$$

The noise amplitude,  $A_{ni}$ , is the magnitude of the value of  $y_0$  that satisfies the relationship  $P(y_0) = DER_0$  where  $DER_0$  is the target detector error ratio. The detector error ratio is the probability that the detector fails to identify the signal level that was transmitted.

In 93A.1.7.1 through 93A.1.7.3, “\*” denotes convolution, which is defined by Equation (93A-38).

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau \quad (93A-38)$$

#### 93A.1.7.1 Interference amplitude distribution

The interference amplitude distribution is computed from the sampled pulse response  $h(n)$  with the assumption that the transmitted symbols are independent, identically distributed random variables and that the symbols are uniformly distributed across the set of  $L$  possible values. For the purpose of this subclause,  $h(n)$  is a general notation that corresponds to  $A_{DD}h_f(n)$  (see 93A.1.7.2),  $h_{ISI}(n)$ , or  $h^{(k)}((i/M+n)T_b)$  (see 93A.1.7.3).

Equation (93A-39) defines the  $n$ th component of the interference amplitude distribution function where  $\delta(y)$  is the Dirac delta function.

$$p_n(y) = \frac{1}{L} \sum_{l=0}^{L-1} \delta\left(y - \left(\frac{2l}{L-1} - 1\right)h(n)\right) \quad (93A-39)$$

The set of  $N$  such components are combined via convolution to obtain the complete interference amplitude distribution. Initialize  $p(y)$  to  $\delta(y)$  and then evaluate Equation (93A-40) sequentially for  $n=0$  to  $N-1$ .

$$p(y) = p(y) * p_n(y) \quad (93A-40)$$

NOTE 1—COM is expected to be numerically computed using a quantized amplitude axis  $y$ . The amplitude step  $\Delta y$  introduces quantization error in the calculated distribution function that is compounded by subsequent convolutions with other quantized distribution functions. It is recommended that  $\Delta y$  be no larger than 0.1% of  $A_s$  or 0.01 mV, whichever is smaller.

NOTE 2—It is recommended that components of the pulse response whose amplitude is less than 0.1% of  $A_s$  be ignored as they likely correspond to measurement noise or numerical artifacts.

### 93A.1.7.2 Noise amplitude distribution

The calculation of COM includes two noise terms that are described in terms of their distribution function. The first term has a Gaussian amplitude distribution function with zero mean and variance  $\sigma_G^2$ . The variance is defined by Equation (93A-41) where  $H_r(f)$  is defined in 93A.1.4.1,  $H_{ctf}(f)$  is defined in 93A.1.4.3, and  $\sigma_X^2$  and  $h_J(n)$  are defined in 93A.1.6.

$$\sigma_G^2 = \sigma_{TX}^2 + \sigma_{RJ}^2 \sigma_X^2 \sum_n h_J^2(n) + \eta_0 \int_0^\infty |H_r(f) H_{ctf}(f)|^2 df \quad (93A-41)$$

The amplitude distribution of the Gaussian noise term is defined by Equation (93A-42).

$$p_G(y) = \frac{\exp(-y^2 / (2\sigma_G^2))}{\sqrt{2\pi\sigma_G^2}} \quad (93A-42)$$

The second term is denoted as  $p_{DD}$  and is related to the amplitude noise resulting from dual-Dirac jitter. It is computed using the procedure defined in 93A.1.7.1 with  $h(n) = A_{DD}h_J(n)$ .

The components are combined using convolution to yield the overall noise amplitude distribution function as defined in Equation (93A-43).

$$p_n(y) = p_G(y) * p_{DD}(y) \quad (93A-43)$$

### 93A.1.7.3 Combination of interference and noise distributions

Compute the intersymbol interference amplitude distribution using the procedure defined in 93A.1.7.1 with  $h(n) = h_{IS}(n)$  as defined by Equation (93A-27) and denote the result as  $p(y)$ .

The contributions of the  $K-1$  crosstalk paths to the total interference are included as follows. Determine the phase index  $m = i$  that maximizes the variance of the amplitude for path  $k$  as defined by Equation (93A-33). Compute the interference amplitude distribution using the procedure defined in 93A.1.7.1 with  $h(n) = h^{(k)}((i / M + n)T_b)$  and denote the result as  $p^{(k)}(y)$ .

Compute  $p^{(k)}(y)$  and evaluate Equation (93A-44) sequentially for integer values  $k=1$  to  $K-1$ .

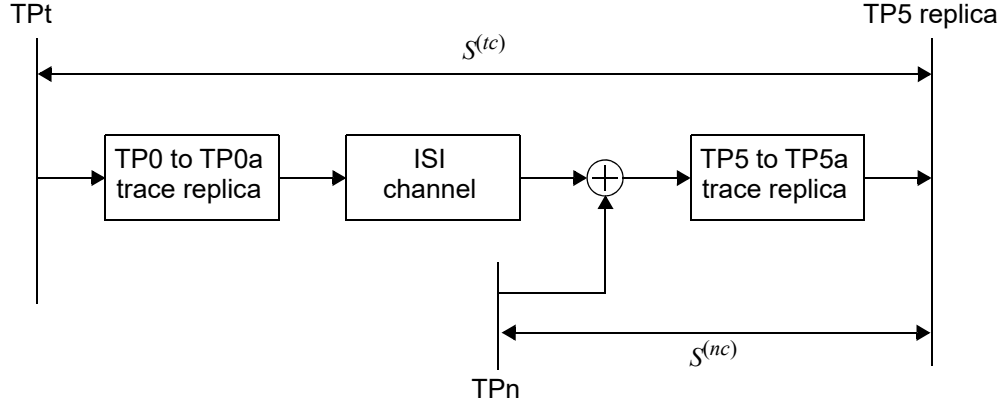
$$p(y) = p(y) * p^{(k)}(y) \quad (93A-44)$$

The noise distribution  $p_n(y)$  defined in 93A.1.7.2 is then included to yield the combined interference and noise amplitude distribution as shown in Equation (93A-45).

$$p(y) = p(y) * p_n(y) \quad (93A-45)$$

## 93A.2 Test channel calibration using COM

A generalized block diagram of the interference tolerance test channel is shown in Figure 93A–2.



**Figure 93A–2—Generalized interference tolerance test channel**

The signal path from the test transmitter connected at TPt to the receiver under test connected at TP5 consists of replicas of the test fixture traces, a controlled ISI channel, and the means by which additive broadband noise is coupled into the path. This path is represented by the scattering parameters  $S^{(tc)}$  measured from TPt to the TP5 replica. The signal path from the broadband noise source connected at TPn to the receiver consists of the means of broadband noise coupling and a replica of TP5 to TP5a test fixture trace. This path is represented by scattering parameters  $S^{(nc)}$  measured from TPn to the TP5 replica.

COM is used to calibrate the interference tolerance test channel. The values assigned to the parameters listed in Table 93A–1 are defined by the Physical Layer specification that invokes this method.

The calculations defined in 93A.1 are evaluated for  $S^{(tc)}$  with the following exceptions.

If the test transmitter presents a high-quality termination, e.g., it is a piece of test equipment, the transmitter device package model  $S^{(tp)}$  is omitted from the calculation of  $S_p^{(k)}$  and the filtered voltage transfer function  $H^{(k)}(f)$  in 93A.1.4 includes the filter  $H_t(f)$  defined by Equation (93A–46) where  $T_r$  is the 20% to 80% transition time in ns (see 86A.5.3.3) of the signal as measured at TP0a and  $f$  is frequency in GHz.

$$H_t(f) = \exp(-2(\pi f T_r / 1.6832)^2) \quad (93A-46)$$

The approximate voltage transfer function for the path from TPn to the output of the receiver equalizer is defined by Equation (93A–47).

$$H^{(ne)}(f) = s_{21}^{(nc)}(f) s_{21}^{(rp)}(f) H_r(f) H_{ctf}(f) \quad (93A-47)$$

The broadband noise source applies noise at TPn that has a Gaussian amplitude distribution with zero mean and standard deviation  $\sigma_{bn}$ . The power spectral density of the noise is flat from  $-f_b/2$  to  $f_b/2$  and is zero elsewhere. The standard deviation of the noise at the receiver equalizer output  $\sigma_{ne}$  is defined by Equation (93A–48).

$$\sigma_{ne}^2 = \frac{2\sigma_{bn}^2}{f_b} \int_0^{f_b/2} |H^{(ne)}(f)|^2 df \quad (93A-48)$$

Equation (93A–41) defines the standard deviation of the Gaussian noise amplitude distribution function. When COM is used to calibrate the interference tolerance test channel, this definition is replaced by Equation (93A–49). The value of  $\sigma_{bn}$  is adjusted until the target COM value is achieved.

$$\sigma_G^2 = \sigma_{TX}^2 + \sigma_{RJ}^2 \sigma_X^2 \sum_n h_J^2(n) + \eta_0 \int_0^\infty |H_r(f) H_{ctf}(f)|^2 df + \sigma_{ne}^2 \quad (93A-49)$$

An additional figure of merit for the test channel is the root-sum-square of the magnitude terms  $n_1$  to  $n_2$  of the equalized pulse response where  $n_2$  is less than or equal to  $N_b$ . This measure of the relative usage of the decision feedback equalizer is defined by Equation (93A–50).

$$u_b(n_1, n_2) = \frac{1}{A_s \sqrt{N}} \sqrt{\sum_{n=n_1}^{n_2} (h^{(0)}(t_s + nT_b))^2} \quad (93A-50)$$

The shorthand notation RSS\_DFE4 is used to represent  $u_b(4, N_b)$ .

### 93A.3 Fitted insertion loss

The fitted insertion loss as a function of frequency is given by Equation (93A–51).

$$IL_{fitted}(f) = a_0 + a_1 \sqrt{f} + a_2 f + a_4 f^2 \quad (93A-51)$$

Denote the insertion loss, in dB, measured at frequency  $f_n$  as  $IL(f_n)$ . Given the insertion loss measured at  $N$  uniformly-spaced frequencies from start frequency  $f_{\min}$  to stop frequency  $f_{\max}$  with step no larger than  $\Delta f$ , the coefficients for the fitted insertion loss shall be calculated as follows.

Define the weighted frequency matrix  $F$  using Equation (93A–52).

$$F = \begin{bmatrix} 10^{-IL(f_1)/20} & \sqrt{f_1} 10^{-IL(f_1)/20} & f_1 10^{-IL(f_1)/20} & f_1^2 10^{-IL(f_1)/20} \\ 10^{-IL(f_2)/20} & \sqrt{f_2} 10^{-IL(f_2)/20} & f_2 10^{-IL(f_2)/20} & f_2^2 10^{-IL(f_2)/20} \\ \dots & \dots & \dots & \dots \\ 10^{-IL(f_N)/20} & \sqrt{f_N} 10^{-IL(f_N)/20} & f_N 10^{-IL(f_N)/20} & f_N^2 10^{-IL(f_N)/20} \end{bmatrix} \quad (93A-52)$$

Define the weighted insertion loss vector  $L$  using Equation (93A–53).

$$L = \begin{bmatrix} IL(f_1) 10^{-IL(f_1)/20} \\ IL(f_2) 10^{-IL(f_2)/20} \\ \dots \\ IL(f_N) 10^{-IL(f_N)/20} \end{bmatrix} \quad (93A-53)$$

The fitted insertion loss coefficients are then given by Equation (93A–54).

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_4 \end{bmatrix} = (F^T F)^{-1} F^T L \quad (93A-54)$$

The values assigned to  $f_{\min}$ ,  $f_{\max}$ , and  $\Delta f$  are defined by the Physical Layer specification that invokes this method.

### 93A.4 Insertion loss deviation

The insertion loss deviation  $ILD(f)$  is the difference between the measured insertion loss  $IL(f)$  and the fitted insertion loss  $IL_{fitted}(f)$  (see 93A.3) as shown in Equation (93A-55).

$$ILD(f) = IL(f) - IL_{fitted}(f) \quad (93A-55)$$

A figure of merit for a channel that is based on  $ILD(f)$  is given by Equation (93A-56). In Equation (93A-56),  $f_n$  are the frequencies considered in the computation of the fitted insertion loss and  $W(f_n)$  is the weight at each frequency as defined by Equation (93A-57).

$$FOM_{ILD} = \left[ \frac{1}{N} \sum_n W(f_n) ILD^2(f_n) \right]^{1/2} \quad (93A-56)$$

$$W(f_n) = \text{sinc}^2(f_n/f_b) \left[ \frac{1}{1 + (f_n/f_t)^4} \right] \left[ \frac{1}{1 + (f_n/f_r)^8} \right] \quad (93A-57)$$

The variable  $f_b$  is the signaling rate. The 3 dB transmit filter bandwidth  $f_t$  is inversely proportional to the 20% to 80% rise and fall time  $T_t$ . The constant of proportionality is 0.2365 (e.g.,  $T_t f_t = 0.2365$ ; with  $f_t$  in Hertz and  $T_t$  in seconds). The variable  $f_r$  is the 3 dB reference receiver bandwidth.

The values assigned to  $f_b$ ,  $T_t$ , and  $f_r$  are defined by the Physical Layer specification that invokes this method.

### 93A.5 Effective Return Loss

Effective Return Loss (ERL) is a figure of merit for the electromagnetic wave reflection from a device or a channel input or output. ERL shall be calculated using the method described in this annex.

The parameters used to calculate ERL are listed in Table 93A-4. The values assigned to these parameters are defined by the Physical Layer specification that invokes the ERL method.

#### 93A.5.1 Pulse time-domain reflection signal

ERL is derived from a unity pulse time-domain reflection signal, PTDR( $t$ ). PTDR( $t$ ) is defined at the test points defined in the Physical Layer specification that invokes the ERL method. PTDR( $t$ ) may be acquired directly from an appropriately filtered time domain reflectometer (TDR), or derived mathematically from measured differential scattering parameters  $S(f)$  and transmitter and receiver filters, according to the procedure in this subclause. See 93A.1.1 for scattering parameters measurement recommendations including frequency step, start frequency, and stop frequency.

**Table 93A-4—ERL parameters**

| Parameter                                                                       | Reference | Symbol    | Units |
|---------------------------------------------------------------------------------|-----------|-----------|-------|
| Signaling rate                                                                  | 93A.1.1   | $f_b$     | GBd   |
| Transition time associated with a pulse                                         | 93A.2     | $T_r$     | ns    |
| Receiver 3 dB bandwidth                                                         | 93A.1.4.1 | $f_r$     | GHz   |
| Number of signal levels                                                         | 93A.1.6   | $L$       | —     |
| Length of the reflection signal                                                 |           | $N$       | UI    |
| Number of samples per unit interval                                             | 93A.1.6   | $M$       | —     |
| Equalizer length associated with reflection signal                              | 93A.5.2   | $N_{bx}$  | UI    |
| Incremental available signal loss factor                                        | 93A.5.2   | $\beta_x$ | GHz   |
| Permitted reflection from a transmission line external to the device under test | 93A.5.2   | $\rho_x$  | —     |
| Target detector error ratio                                                     | 93A.1.7   | $DER_0$   | —     |

The filtered return loss,  $H_{ii}(f)$ , is defined by Equation (93A-58).

$$H_{ii}(f) = H_i(f)s_{ii}(f)H_r(f) \quad (93A-58)$$

where

- $f$  is the frequency in GHz
- $H_r(f)$  is defined by Equation (93A-20)
- $H_i(f)$  is defined by Equation (93A-46)
- $i$  is the port index of the scattering parameters, 1 or 2

The pulse TDR signal,  $PTDR(t)$ , is defined by Equation (93A-59).

$$PTDR(t) = \int_{-\infty}^{\infty} X(f)H_{ii}(f)\exp(j2\pi ft)df \quad (93A-59)$$

where

- $t$  is the time in ns starting from the peak of the injected pulse
- $X(f)$  is defined by Equation (93A-23) with  $A_t$  set to 1

### 93A.5.2 Effective reflection waveform

The effective reflection waveform,  $R_{eff}(t)$ , is computed by time gating and weighting the PTDR waveform,  $PTDR(t)$ , according to Equation (93A-60).  $R_{eff}(t)$  is a pure number.

$$R_{eff}(t) = PTDR(t) \times G_{rr}(t) \times G_{loss}(t) \quad (93A-60)$$

where  $G_{rr}(t)$  and  $G_{loss}(t)$  are time gating weighting functions defined in Equation (93A–61) and Equation (93A–62) with  $t$  in nanoseconds.

$$G_{rr}(t) = \begin{cases} 0 & t < T_{fx} \\ \rho_x(1 + \rho_x) \exp\left(-\frac{[(t - T_{fx})f_b - (N_{bx} + 1)]^2}{(N_{bx} + 1)^2}\right) & T_{fx} \leq t < T_{fx} + \frac{N_{bx} + 1}{f_b} \\ 1 & t \geq T_{fx} + \frac{N_{bx} + 1}{f_b} \end{cases} \quad (93A-61)$$

where

$t$  is the time in ns starting from the peak of the injected pulse  
 $T_{fx}$  is twice the propagation delay in ns associated with the test fixture, obtained by measurement or inspection  
 $\rho_x, f_b, N_{bx}$  are supplied by the clause that invokes this method

$$G_{loss}(t) = \begin{cases} 0 & t < T_{fx} \\ 10^{\frac{\beta_x[(t - T_{fx})f_b - (N_{bx} + 1)]}{20}} & T_{fx} \leq t < T_{fx} + \frac{N_{bx} + 1}{f_b} \\ 1 & t \geq T_{fx} + \frac{N_{bx} + 1}{f_b} \end{cases} \quad (93A-62)$$

where

$t$  is the time in ns starting from the peak of the injected pulse  
 $T_{fx}$  is twice the propagation delay in ns associated with the test fixture, obtained by measurement or inspection  
 $\beta_x, f_b, N_{bx}$  are supplied by the clause that invokes this method

### 93A.5.3 Sampled effective reflection

The sampled effective reflection for each phase  $m$  is computed per Equation (93A–63).

$$h^{(m)}(n) = R_{eff}\left(T_{fx} + \frac{n + m/M}{f_b}\right) \quad (93A-63)$$

where

$n$  is an integer ranging from 0 to  $N-1$   
 $m$  is an integer ranging from 0 to  $M-1$   
 $N$  and  $M$  are supplied by the clause that invokes this method

The standard deviation of the distribution of the reflection signal for each phase  $m$ ,  $\sigma^{(m)}_h$ , is defined by Equation (93A–64).

$$\sigma^{(m)}_h = \sqrt{\sum_{n=1}^N h^{(m)}(n)^2} \quad (93A-64)$$

where

$m$  is an integer ranging from 0 to  $M-1$   
 $N$  and  $M$  are supplied by the clause that invokes this method

#### 93A.5.4 x-quantile of the reflection distribution

The reflection signal distribution  $p(y)$  is computed from the sampled effective reflection using the procedure defined in 93A.1.7.1, with  $h(n) = h^{(m)}(n)$ , where  $m$  maximizes  $\sigma^{(m)}_h$ . The value of  $L$  in Equation (93A-39) is supplied by the clause that invokes this method. The corresponding cumulative distribution function  $P(y)$  is calculated from  $p(y)$  using Equation (93A-37).

The  $x$ -quantile of the distribution,  $P^{-1}(x)$ , is the value of  $y$  that satisfies the relationship  $P(y)=x$ .

#### 93A.5.5 ERL

ERL is defined as  $-20 \times \log_{10} P^{-1}(DER_0)$  where  $DER_0$  is the target detector error ratio.